

NARAYANAN PETCHIMUTHU
PRINCIPAL FULL STACK SOFTWARE ENGINEER

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- *Principal Generative AI Developer & Technical Architect* with extensive hands-on experience designing, scaling, and deploying production-grade enterprise software solutions.
- *Expertise in Python & Generative AI:* Proven track record employing Azure AI Foundry, Azure OpenAI SDK, and vector embeddings to architect high-precision RAG pipelines, multi-agent workflows, and conversational memory systems.
- *Enterprise Full-Stack Architecture:* Extensive background developing resilient microservices and distributed applications using C#, .NET, Node.js, Express, and modern web APIs.
- *Database & Data Engineering Mastery:* Deep mastery of relational and NoSQL database management, including complex query and schema optimization across SQL Server, Oracle, CosmosDB, and SQLite.
- *Cloud Native & DevOps Execution:* Hands-on Azure Cloud Practitioner skilled in automated CI/CD pipelines, cloud hosting, and containerized deployment environments using Docker and Kubernetes.
- *Leadership & Cross-Functional Impact:* Distinguished career as Architect, Technical Lead, and Client Liaison, driving cross-functional delivery and mentoring engineering teams across Retail, Energy, Healthcare, Media and Financial Services.
- *Industry Credentials & Certifications:*
 - *Google AI Professional / Google Project Management:* Specialization
 - *Microsoft Certified Professional:* Azure Fundamentals / Azure AI Fundamentals / Designing & Implementing Databases with MS SQL Server Enterprise

SKILLS

• Architecture & Distributed Systems	<i>System Design, Asynchronous & Concurrent programming, Event-Driven Architecture, Azure Service Bus, Microservices, RAG Architecture</i>
• Cloud & DevOps	<i>Azure Functions, Azure API Manager (APIM), Azure Redis, Azure DevOps, Github, AWS, CI/CD Pipelines, Docker, Kubernetes, App Insights, Dynatrace, Splunk</i>
• Generative AI & LLMs	<i>Python, Azure OpenAI SDK, OpenAI APIs, Azure AI Foundry, Google ADK, LangChain, LangGraph, mlflow (LLMOps), Amazon Bedrock, Prompt Flow, FastMCP, Copilot</i>
• Data Management	<i>Azure AI Vector Index, ChromaDB, Azure Data Factory (ADF), Redis, SQL Server, Oracle, CosmosDB, MongoDB, SQLite, ElasticSearch, SSIS</i>
• Scripting & Tools	<i>C#, Asp.Net, Blazor, Entity Framework, MVC, REST, Web API, React, NodeJS, Express, FastAPI, Flask, TypeScript, HTML/CSS, OAuth, Swagger, Apache JMeter</i>
• Project Management	<i>Agile, Azure Boards, Scrum, Jira, Remedy, ServiceNow</i>

EXPERIENCE

Meijer Inc. / Randstad (Contract)	Technical Lead & Architect	Apr '20 – Present
• Architected production-grade <i>GenAI</i> solutions in <i>Python</i> leveraging <i>Azure OpenAI</i> and <i>Azure AI Vector Search</i> to deliver high-accuracy <i>Retrieval-Augmented Generation (RAG)</i> over complex datasets. • Enhanced document retrieval precision by implementing structure-aware data chunking alongside a hybrid <i>BM25</i> lexical and <i>semantic vector</i> search model utilizing <i>Reciprocal Rank Fusion (RRF)</i>. • Engineered autonomous AI agents using <i>Python</i> and <i>FastMCP</i>, enabling tool-use capabilities, reasoning loops, and multi-system enterprise API integrations. • Accelerated response times and reduced LLM token costs by implementing <i>prompt caching</i> and <i>context-aware</i> content processing, while integrating robust input/output <i>guardrails</i> to secure sensitive enterprise data and ensure AI output safety. • Containerized and deployed resilient, event-driven microservices using <i>Docker</i> and <i>Kubernetes (AKS)</i>, establishing low-latency, scalable infrastructure for production AI workloads. • Designed backwards-compatible <i>C# .Net Web APIs</i> via <i>Swagger/Azure APIM</i> and built <i>Azure Data Factory (ADF)</i> pipelines to		

ingest Blob data to *Cosmos DB* and trigger *Azure Service Bus* messaging.

- Boosted response times by 38% using *Task Parallel Library* async/multithreading, implemented low-latency *Azure Redis* caching for product/order data, and benchmarked endpoints via *NUnit* and *Apache JMeter*.
- Implemented *Okta OAuth 2.0* SSO, developed *Azure Functions* for third-party SAAS API risk assessments (*Sayari*), maintained code quality via *Aikido* vulnerability scans and *GitHub Copilot* reviews.
- Managed CI/CD release pipelines using *GitHub* and *Azure DevOps*, while actively monitoring application health, availability, and cost via *Application Insights*, *Dynatrace*, and *Splunk*.

NRG Energy / Randstad (Contract)

Lead Developer & Architect

Nov '14 – Dec '19

- Architected and maintained full-stack customer account web portals using *Asp.Net MVC*, *Ajax*, *jQuery*, and *AngularJS*.
- Engineered SOA layer components in *C# (WCF)* to integrate CRM data services via *REST APIs*.
- Optimized application memory and load throughput by implementing *Memcached* memory management and *Blowery HTTP* compression.
- Designed and integrated embedded *SSRS* reports to deliver dynamic customer energy usage insights directly onto portal UI pages.
- Developed database schemas, stored procedures, and an *Entity Framework* Data Access Layer (DAL) for robust *CRUD* operations in *SQL Server 2012*.
- Configured automated application build scripts and scheduled daily *SQL Server* maintenance and processing jobs.

Halliburton / Randstad (Contract)

Senior Developer

Mar' 12 – Oct' 14

- Architected *C# WCF* service layer components and defined contracts to stream field data seamlessly across diverse business workstreams.
- Designed custom serialization routines utilizing *Xml*, *Binary*, and *DataContract* serializers for optimized data exchange across services.
- Integrated *RabbitMQ* message queuing to ensure resilient, decoupled client-server data transfer.
- Secured service endpoints by implementing *X509 certificate*-based authorization over *net.tcp* protocols.
- Optimized network throughput by 2.5x and reduced payload size by 60% by implementing *Protocol Buffers* for fast binary serialization.
- Developed *SQL Server* database schemas, stored procedures, and caching jobs to ingest and process real-time and historical field data.

Kinder Morgan / Randstad (Contract)

Lead Developer

Apr' 11 – Mar' 12

- Engineered custom Windows utilities using *Profisee Maestro Master Data Services* (MDS) APIs to automate environment migrations, model package exports, and security administration.
- Developed SOA layer components in *C# (WCF)* integrating MDS APIs to define contracts and stream master data across enterprise systems.
- Authored *T-SQL* queries and stored procedures in *SQL Server 2008* to execute backend data logic for custom MDS workflows.
- Built *SSIS ETL* pipelines to staging and master data environments, designing corresponding *SSRS* reporting dashboards.
- Customized integration portal UI styling using *CSS* for consistent cross-application design.

Deloitte

Associate

Apr '09 – Apr '11

- Engineered backend database logic in *Oracle DB* using *PL/SQL* queries, packages, and stored procedures.
- Developed custom *C#* and *VB.NET* script modules to support core application features and complex data conversion workflows.
- Architected *C# WCF* service layer components to manage logging mechanisms, screening business rules, and external enterprise system interfaces.
- Designed *SSIS ETL* packages to automate legacy flat-file data ingestion into *SQL Server 2005* staging tables.
- Built front-end UI components for the client self-service portal using *Adobe Flex Builder 3*.
- *Key Recognition*: Honored with '*Outstanding Performance Award*' by Deloitte and client leadership (CDHS) for exceptional technical contributions towards successful rollout of the Child Care Accounting & Tracking System (CHATS) platform.

EDUCATION

Bachelor of Engineering in Computer Science & Engineering, Madurai Kamaraj University, India [4.0 GPA]